Record-breaking magnet
A superconducting magnet beat the world record by sustaining a 45.5 tesla magnetic field intensity. **https://dgit.in/july19-12**

Raspberry Pi 4 packs a punch
The Raspberry Pi 4 can output 4K videos at 60fps, and sports better Bluetooth and USB connectivity. **https://dgit.in/july19-13**

The Chaos Pendulum

Magnets are fun

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There are a number of forces acting on the pendulum. They includes gravity, the magnets on the metal panel, and air resistance, or friction. Normally, the movement of a pendulum is very regular and predictable, which allows them to be used in precision instruments such as grandfather clocks. However, introduce magnets beneath a magnetic pendulum, and you get chaos. This DIY works great as a desk toy that you can fiddle around with in your home or office.

STEP 1

For this build, we will require the following materials. Once you get the basic idea, you can easily modify the build according to the resources on hand.

Materials:

- Sticky Tape
- Plasticine
- Magnets (more than 4)
- String
- Copper wire
- Foam
- Metal Plate

Tools:

- Glue Gun (can be replaced with any strong glue)
- Scissors
- Thin screwdriver or needle

STEP 2

Cut a small piece of foam. This can be circular or square,

it doesn't really matter. It should be roughly the size of the magnet. You will have to make sure the foam is at least 1 cm in thickness.

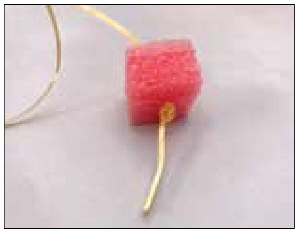
STEP 3

The string will be passed through this. You can use a screwdriver or a refill or any pointy thing at hand. Just

make sure that the diameter of the hole is not too much bigger than the string.

STEP 4

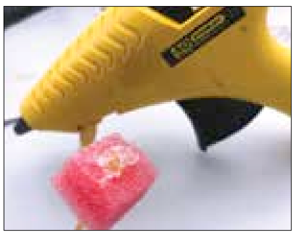
Tie a couple of knots at the end of the string, and pass it through the piece of foam.



You will need to figure out an approximate length of the string, depending on where you are attaching it. We used an overhanging shelf, but you can make a short stand as well. Leave some headroom in terms of making it longer, so that you can cut off the excess string while making the final adjustments. Cut off the string after the knot.

STEP 5

Now heat up the glue gun, and daub on hot glue to the



knot at the end of the foam. The magnet can be attached here. Using superglue is better, but if anything goes wrong, the hot glue can simply be peeled off after it cools, the same cannot be done with superglue.

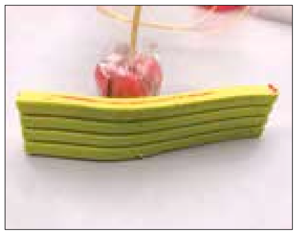
STEP 6

Whether you use superglue or hot glue, you will need to additionally secure the magnet. We used copper wire and sticky tape. Rubber bands are also a good

idea. The magnet needs to be so firmly fixed, because magnetism is a really, really strong force. The gravity of the entire Earth cannot make the fridge magnet fall. Think about that. If you do not secure the magnet sufficiently, it will escape from the foam base within a few days. Bear in mind that the metal plate will be constantly pulling the magnet downwards.

STEP 7

Now you need to make a



thin strip of clay, and wrap it around the magnet and foam piece. The bottom of the magnet can remain exposed. We chose clay as it is the easiest way to add weight to the pendulum, but you can use any other method as well. The basic idea is to make the pendulum heavier.

STEP 8

The clay is wrapped around the pendulum. Make sure that the weight is evenly distributed, and the pendulum